Zishuo Huang

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Education

Department of Electrical Engineering, Tsinghua University — Beijing, China

Ph.D. in Electrical Engineering — 2022 - 2027 *(expected)*

- Advisor: Prof. Wenchuan Wu
- Research area: optimization and control of renewable energy sources, especially wind farms

School of Electrical Engineering, Xi'an Jiaotong University — Xi'an, China

B.E. in Electrical Engineering (Qian Xuesen's Class) — 2018 - 2022

Class of the Gifted Young, Xi'an Jiaotong University — Xi'an, China

2016 - 2018

Publications

1. **Zishuo Huang**, Chenhui Lin, Qi Wang, Wenchuan Wu. Frequency-Routed Coordinated Pitch-Torque Control for Fatigue-Relevant Response Mitigation in Floating Offshore Wind Turbines. *IEEE Transactions on Sustainable Energy*, Early Access, 2026.(**IF: 10.0**)
2. **Zishuo Huang**, Chenhui Lin, Qi Wang, Wenchuan Wu. A Bilevel Optimization Method for Wind Farm Layout Considering Yaw Control. *IEEE Transactions on Industrial Informatics*, 2025. (**IF: 9.9**)
3. **Zishuo Huang**, Wenchuan Wu. Dual-Stage MPC-Based AGC for Wind Farm Considering Aerodynamic Interactions. *IEEE Transactions on Sustainable Energy*, 2025. (**IF: 10.0**)
4. **Zishuo Huang**, Wenchuan Wu, Chenhui Lin, Zizhen Guo. Cooperative Strategies for Frequency Control of Wind Turbines to Mitigate Secondary Frequency Dip: Coefficient Allocation and Exit Techniques. *IEEE Transactions on Sustainable Energy*, 2025. (**IF: 10.0**)
5. **Zishuo Huang**, Wenchuan Wu. An Efficient Solution for Large Offshore Wind Farm Power Optimization with the Porté-Agel Wake Model: Optimality and Efficiency. *Energy*, 2024. (**IF: 9.4**)
6. **Zishuo Huang**, Wenchuan Wu, Zizhen Guo. Data-driven Stochastic Model Predictive Control for Regulating Wind Farm Power Generation with Controlled Battery Storage. *IET Renewable Power Generation*, 2023. (**IF: 2.9**)
7. **Zishuo Huang**, Wenchuan Wu, Hongyan Liu, Jisi Wu, Hao Liu. Real-Time Digital Simulation of DFIG Wind Farm Primary Frequency Control via Droop and Virtual Inertia Control. *6th International Conference on Electrical Technology and Automatic Control (ICETAC)*, 2025.
8. Guojian Li, **Zishuo Huang**, Zhenfeng Wang, Jin Xu, Jihui Dong, Jiang Li. Cooperative Yaw and Induction Control for Wind Farm and Its Effectiveness in Providing Secondary Frequency Regulation. *IEEE 8th Conference on Energy Internet and Energy System Integration (EI2)*, 2024.
9. Guannan Wang, **Zishuo Huang**, Wenchuan Wu, Xiaowei Fan, Jun Zhang, Haitao Liu. Model Predictive Control for Regulating Battery Storage to Smooth the Wind Farm Power Generation. *IEEE International Conference on Power System Technology*, 2023.

Projects

Zhangjiakou 100% Renewable Energy Aggregation and Operational Control: Research & Demonstration

Tsinghua University 2023.04 - 2025.07

In this project, I was responsible for modeling and implementing a plant-level active support control scheme. I first proposed a measurement-driven lifted-mapping approach for global linearization, mapping low-dimensional, highly nonlinear processes to a high-dimensional linear state-space model, which produced an interpretable and tractable dynamic-control prototype and a model–data co-driven framework for power tracking and frequency/voltage support; I then extended the prototype to a **stochastic MPC** with explicit wind-speed forecast-error correction and a deterministic reformulation, unifying in the objective function the penalties on grid-side power fluctuation, schedule-tracking deviation, curtailment, and **battery degradation cost**, while enforcing engineering constraints on **energy storage, pitch angle, and grid voltage** to realize wind–storage coordinated optimization. The control performance of the cluster controller was validated via real-time simulation on RTDS.

Awards and Honors

- **National Second Prize (Team Leader)**, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2019.11
- **Provincial First Prize (Shaanxi)**, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2018.11
- **First Prize (Shaanxi Division)**, 11th National Undergraduate Mathematics Competition 2019.10
- **First Prize**, Xi'an Jiaotong University Mathematical Modeling Competition 2019.05
- **First Prize**, Xi'an Jiaotong University Mathematical Modeling Competition 2018.05
- **Qiu Changrong Scholarship (Second Class)**, Academic Year 2019–2020 2020.11
- **Qiu Changrong Scholarship (Second Class)**, Academic Year 2018–2019 2019.11
- **Outstanding Student**, Qian Xuesen College, Academic Year 2019–2020 2020.10
- **Outstanding Student**, Qian Xuesen College, Academic Year 2018–2019 2019.10

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Skills

Languages: Chinese, English.

Programming: `Python`, `C++`, `MATLAB`.

Simulation softwares: `Simulink`, `RTDS(RSCAD)`, `Powerfactory`

Academic Services

Reviewers for: *IEEE Transactions on Sustainable Energy*,
IEEE Transactions on Industrial Informatics,
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